

Building and Evaluating Recommender Systems

Final Project



Capstone Machine Learning

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Introduction

Recommender systems help users discover relevant items efficiently

This project focuses on course recommendation systems to help learners discover relevant courses

■ Project Objectives

- Build content-based and collaborative filtering models
- Compare their performance

■ Techniques Include



Similarity-Based

Methods that measure item similarity



Matrix Factorization

Decomposes user-item interactions



Neural Networks

Learning non-linear patterns

Dataset Overview

The dataset consists of user-course interaction data for recommendation modeling

User–Course Interaction

Records of user enrollments and activities



Course Metadata

Course genres and categories for content-based filtering



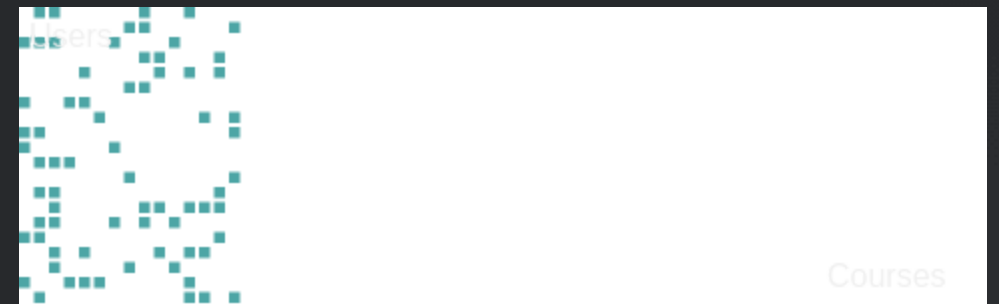
User Ratings

Preferences represented as rating values



Data Sparsity

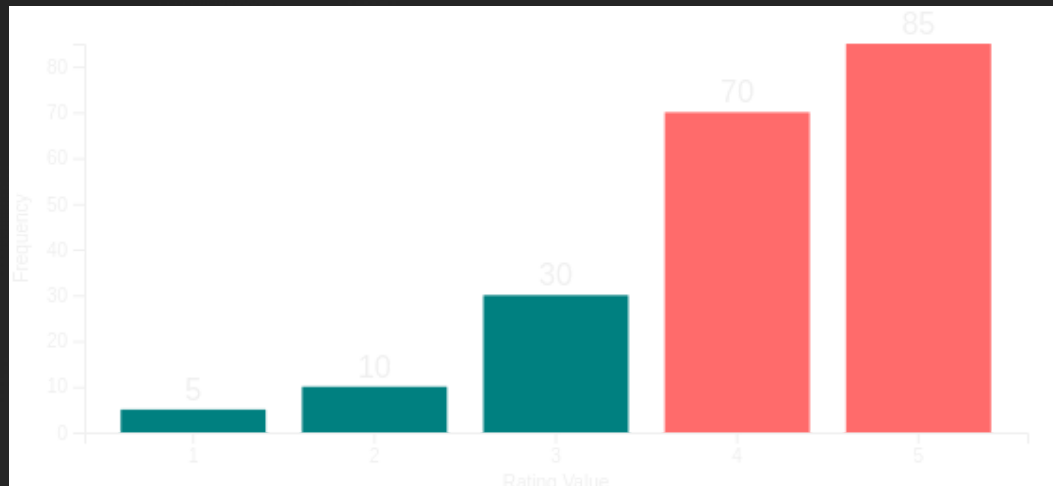
Many users rate few courses



Exploratory Data Analysis - Ratings Distribution

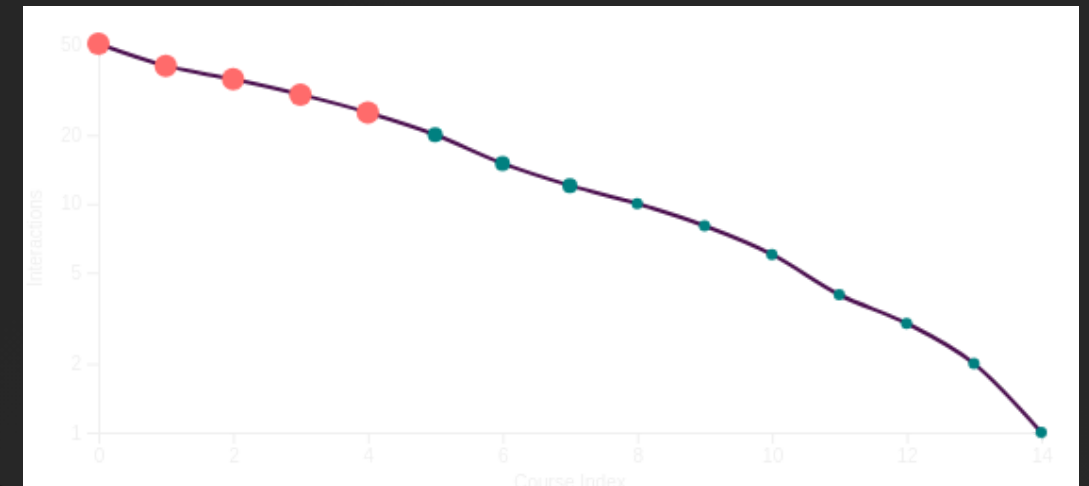
Analysis of rating patterns shows two important characteristics:

Rating Distribution



- Ratings cluster around mid to high values
- Most ratings are 3.0 and above

Long-tail Effect



- Majority of courses receive few interactions
- Popular courses are rare

Key Insights

- Most ratings are 4.0-5.0, indicating positive user experience
- Data is sparse - many users rate few courses

EDA - User and Course Activity Patterns

■ User Activity Variations

- User activity varies significantly
- Some users are highly active



Low Activity



Medium Activity



High Activity



New Users

Cold-start problem exists for new users with limited interaction history



New Courses

New courses lack sufficient interaction data for accurate recommendations



Key Insight

Confirms need for hybrid recommendation approaches that can handle both user and course cold-start scenarios

Content-Based Recommender - User Profile Approach

User profiles are built from genre preferences and represented as weighted vectors

User Profile

User Preferences Vector

Action		0.7
Comedy	0.4	
Drama		0.8
Sci-Fi	0.5	

Match

Course Recommendations

Course Genre Vectors

Action		0.65
Comedy	0.35	
Drama		0.75
Sci-Fi		0.55

Process

- Build user profile from rated courses
- Represent user as genre-weighted vector
- Match with course genre vectors

Advantage

- Works without other users' data
- Simple to implement and understand
- Provides explainable recommendations

Content-Based Recommender - Course Similarity

TF-IDF Representation

Courses represented using TF-IDF vectors based on genre frequencies

TF-IDF weights capture genre importance

Cosine Similarity

Measures angle between course vectors to determine similarity

Value range: 0 (orthogonal) to 1 (identical)

Recommendation Process

- Compare courses with user history
- Recommend most similar courses

Vector Space Visualization



Key Advantages



Effective with history



No user data needed

Content-Based Recommender - User Clustering

Clustering Approach

Users are clustered based on their genre preferences, grouping similar users together to improve recommendations.

Key Components

- Cluster users by genre preferences
- Group similar users together
- Recommend popular courses within clusters

Benefits

- Reduces sparsity problem
- Improves recommendation accuracy

User Clusters Based on Genre Preferences



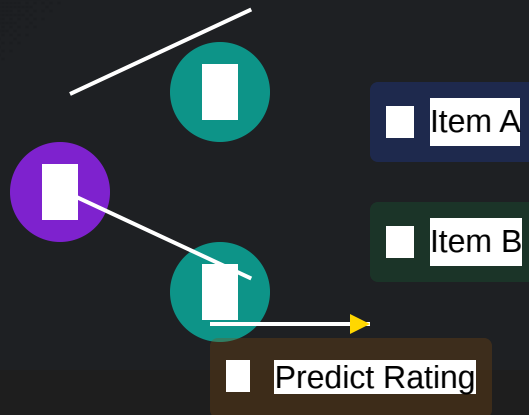
- Clustering helps identify groups of users with similar interests, enabling more accurate course recommendations.

Collaborative Filtering - KNN Approach

User-based KNN collaborative filtering

Finds users with similar rating behavior to predict ratings

How KNN Works



Key Concepts

- Finds users with similar rating patterns
- Uses neighbors' ratings for prediction

■ **Strength:**
Simple and intuitive

■ **Weakness:**
Scalability issues

Collaborative Filtering - Matrix Factorization

Non-negative Matrix Factorization

Decomposes user-item matrix into latent factors

Key Features:



Captures Hidden Preferences
Reveals underlying patterns



Sparse Data Handling
Effective on sparse datasets



Advantages



More scalable than KNN

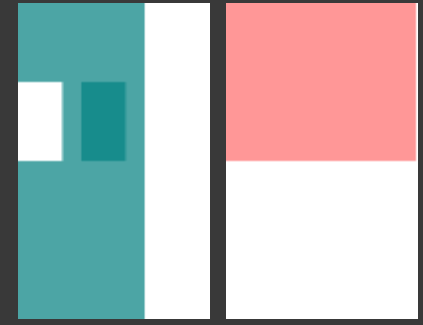
Matrix Factorization Visualization

User-Item Matrix
(R)

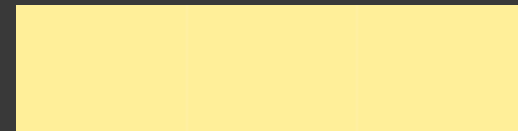


Factorization

User-Factors (P) × Item-Factors (Q^T)



Reconstruction (P × Q^T)



Neural Network Embedding CF

Uses embeddings to learn non-linear interactions between users and courses

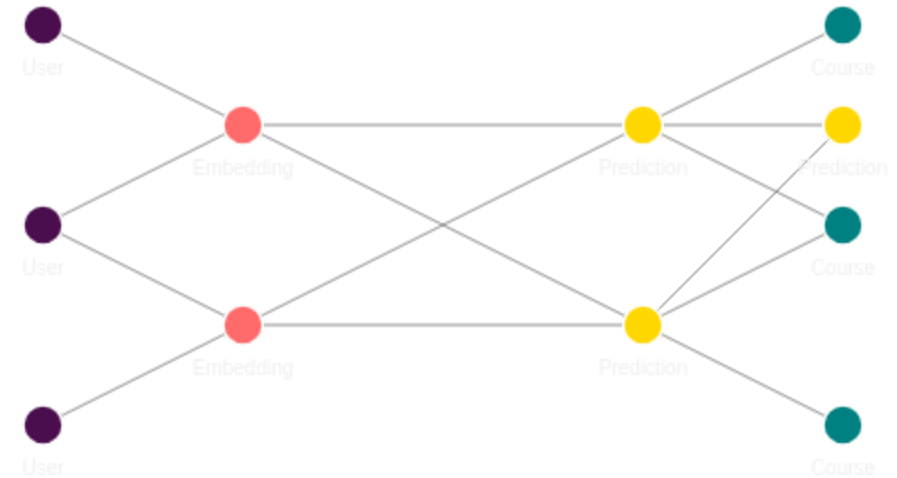
Model Structure

- Embedding layers - Learn latent representations
- Dense layers - Capture interactions

Advantages

- More accurate predictions
- Captures complex patterns

Neural Network Architecture



Users Courses Embeddings Output

Model Evaluation and Performance Comparison

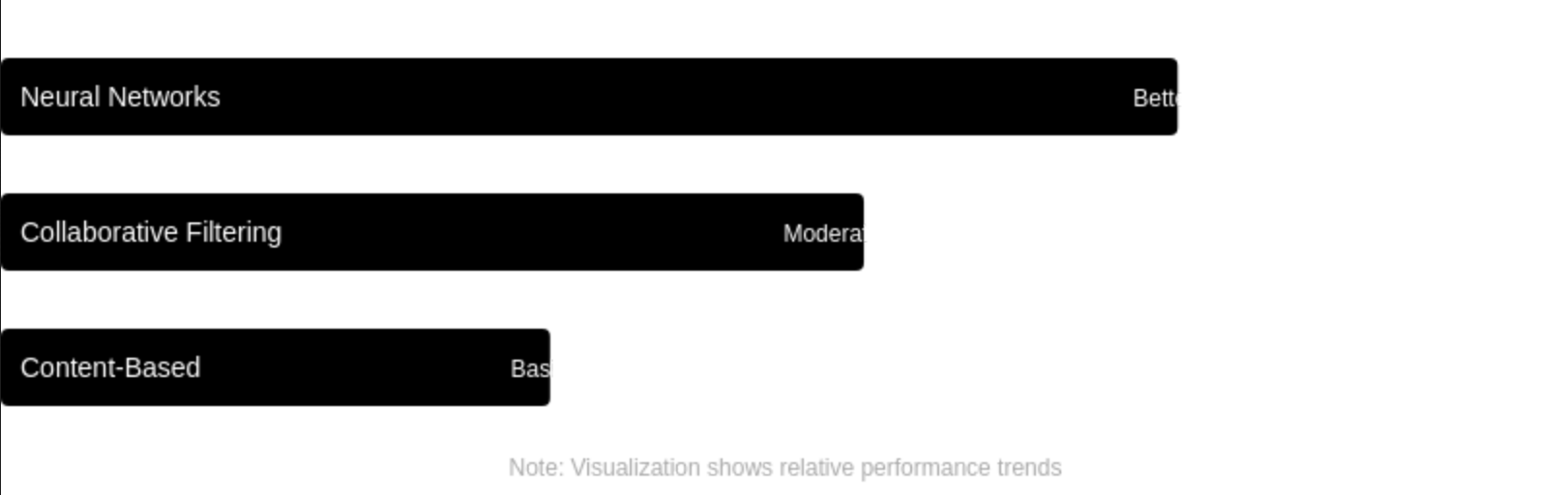
 RMSE

Root Mean Square Error - Lower values indicate better fit

 MAE

Mean Absolute Error - Average prediction error

Relative Model Performance



 Neural Networks  Collaborative Filtering  Content-Based

Key Findings

-  Neural network embeddings perform best
-  Collaborative filtering improves accuracy

Complexity-Performance Trade-off

More complex models don't always outperform simpler ones

Conclusion and Key Findings

Implemented Systems

Multiple recommender systems were implemented:

Content-based (User Profile)

Content-based (Course Similarity)

Content-based (User Clustering)

Collaborative (KNN)

Collaborative (NMF)

Neural Network Embeddings

Performance Comparison

Content-based Methods

Perform reasonably well but limited by user history coverage

Collaborative Filtering

Outperformed content-based methods with better accuracy

Most Promising

Hybrid Approaches

Integration of multiple methods shows superior performance

Creativity, Innovation and Future Work



Technical Integration

- Compared multiple recommendation techniques
- Integrated clustering and embeddings



Real-world Challenges

- Addressed data sparsity issues
- Solved cold-start problems



Future Directions



Hybrid Ensemble Models

Combine multiple approaches for optimal performance



Real-time Personalization

Adaptive systems that respond to user behavior

Thank you for your attention!